Stochastic Interest Rate Blending Weights

Derivation, Dynamics, and Implications for International Finance and Machine Learning

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Abstract

We study a blending equation that expresses the international interest rate r_I as a convex-combination-like mixture of the domestic rate r_D and a spread term, parameterised by a scalar weight w. Starting from a purely algebraic derivation of w in terms of r_D and r_I, we extend both rates to continuous-time stochastic processes modelled as Geometric Brownian Motions (GBM), derive the Itô SDE satisfied by $r_I(t)$, and analyse the pathwise behaviour and singularity structure of $w(t)$. We further connect the framework to uncovered interest parity (UIP), no-arbitrage pricing, and modern machine learning calibration methods. A Monte Carlo simulation algorithm is provided alongside numerical results, parameter tables, and PGF/TikZ illustrations.

The paper ends with "The End"

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1 Introduction

The relationship between domestic and international interest rates is a cornerstone of open-economy macroeconomics, international finance, and fixed-income asset pricing [1, 2]. A natural question is how a single scalar weight w can encode the relative contribution of the domestic rate versus a spread component when expressing r_I as a blended quantity.

The blending equation we study is

$$r_I = w\, r_D + (1 - w)(r_I - r_D), \tag{1}$$

which can be viewed as a *generalised convex combination* where the second argument is the interest-rate spread $r_I - r_D$ rather than r_I itself. Solving for w closes the equation and yields a compact ratio whose properties differ markedly from a standard mixing weight.

In the stochastic setting, both r_D and r_I become continuous-time processes, and $w(t)$ inherits their randomness through a nonlinear map. Understanding the distribution, moments, and tail behaviour of $w(t)$ has practical implications for:

- **Central bank policy**: calibrating exposure to global rate shocks;
- **Fixed-income derivatives**: pricing spread options and quanto products;
- **Machine learning**: using $w(t)$ as a latent factor in neural SDE or Gaussian Process models of the yield curve.

The remainder of the paper is organised as follows. Section 2 derives w algebraically. Section 3 promotes the rates to stochastic processes and derives the SDE for $r_I(t)$. Section 4 analyses the properties of $w(t)$. Section 5 discusses economic interpretations. Section 6 connects the framework to machine learning. Section 7 presents a simulation algorithm. Section 8 reports numerical results. Section 10 concludes.

2 Algebraic Derivation of the Weight

Proposition 2.1 (Closed-Form Weight)**.** *Given the blending equation* (1) *with* $2r_D \neq r_I$*, the unique solution for* w *is*

$$\boxed{w = \frac{r_D}{2r_D - r_I}.} \tag{2}$$

Proof. Expand the right-hand side of (1):

$$\begin{aligned} r_I &= w\, r_D + (1-w)(r_I - r_D) \\ &= w\, r_D + r_I - r_D - w r_I + w r_D \\ &= r_I - r_D + w(2r_D - r_I). \end{aligned} \tag{3}$$

Subtracting r_I from both sides of (3):

$$0 = -r_D + w(2r_D - r_I).$$

Solving for w (assuming $2r_D \neq r_I$):

$$w = \frac{r_D}{2r_D - r_I}.$$

□

Remark 2.2 (Singularity Condition)**.** The weight w is undefined—equivalently, (1) has no unique solution—when $r_I = 2r_D$, i.e. when the international rate is exactly twice the domestic rate.

Remark 2.3 (Convexity)**.** Unlike a standard mixing weight, w need not lie in $[0, 1]$. Specifically:

- $w \in (0, 1)$ iff r_D and r_I share the same sign and $r_I < r_D$;
- $w > 1$ when $0 < r_I < r_D$ and $2r_D - r_I < r_D$, i.e. $r_I > r_D$;
- $w < 0$ is possible for negative spreads.

Table 1 summarises representative parameter regimes.

Table 1: Values of w for selected rate configurations.

r_D (%)	r_I (%)	$2r_D - r_I$ (%)	w
5.0	3.0	7.0	0.714
5.0	6.0	4.0	1.250
5.0	8.0	2.0	2.500
5.0	10.0	0.0	$\pm\infty$
5.0	12.0	−2.0	−2.500
3.0	1.5	4.5	0.667

3 Stochastic Extension

3.1 Specification of the Primitive Processes

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions, and let $B_1(t)$, $B_2(t)$ be standard $\mathcal{F}_t$-Brownian motions with instantaneous correlation

$$\mathrm{d}B_1(t)\,\mathrm{d}B_2(t) = \rho\,\mathrm{d}t, \quad \rho \in (-1, 1).$$

Definition 3.1 (Primitive GBM Processes)**.** We model the domestic rate and blending weight as:

$$\mathrm{d}r_D(t) = \mu_D\, r_D(t)\,\mathrm{d}t + \sigma_D\, r_D(t)\,\mathrm{d}B_1(t), \tag{4}$$

$$\mathrm{d}w(t) = \mu_w\, w(t)\,\mathrm{d}t + \sigma_w\, w(t)\,\mathrm{d}B_2(t), \tag{5}$$

with deterministic initial conditions $r_D(0) = r_0 > 0$ and $w(0) = w_0 > 0$.

Both processes have explicit solutions:

$$r_D(t) = r_0 \exp\left[\left(\mu_D - \tfrac{\sigma_D^2}{2}\right) t + \sigma_D B_1(t)\right], \tag{6}$$

$$w(t) = w_0 \exp\left[\left(\mu_w - \tfrac{\sigma_w^2}{2}\right) t + \sigma_w B_2(t)\right]. \tag{7}$$

3.2 Derived Process for $r_I(t)$

From Proposition 2.1, rearranging $w = r_D/(2r_D - r_I)$ gives

$$r_I(t) = r_D(t)\left(2 - \frac{1}{w(t)}\right) =: f(r_D(t), w(t)). \tag{8}$$

Theorem 3.2 (SDE for the International Rate)**.** *Let $r_D(t)$ and $w(t)$ follow (4)–(5). Define $r_I(t)$ by (8). Then $r_I(t)$ satisfies*

$$\begin{aligned} \mathrm{d}r_I(t) = &\left[\left(\mu_D + \frac{\sigma_w^2}{w^2} - \mu_w\right) r_I(t) + \left(\mu_w - \frac{\sigma_w^2}{w^2} - \rho\,\sigma_D\sigma_w\right)\frac{r_D(t)}{w(t)}\right]\mathrm{d}t \\ &+ \sigma_D\left(2 - \frac{1}{w(t)}\right) r_D(t)\,\mathrm{d}B_1(t) + \frac{\sigma_w\, r_D(t)}{w(t)}\,\mathrm{d}B_2(t). \end{aligned} \tag{9}$$

Proof. Apply Itô's formula to $f(x, y) = x(2 - y^{-1})$ where $x = r_D$, $y = w$:

$$\mathrm{d}f = f_x\,\mathrm{d}x + f_y\,\mathrm{d}y + \tfrac{1}{2}f_{xx}(\mathrm{d}x)^2 + f_{xy}\,\mathrm{d}x\,\mathrm{d}y + \tfrac{1}{2}f_{yy}(\mathrm{d}y)^2. \tag{10}$$

Compute the partial derivatives:

$$f_x = 2 - y^{-1}, \quad f_y = xy^{-2}, \quad f_{xx} = 0, \quad f_{xy} = y^{-2}, \quad f_{yy} = -2xy^{-3}.$$

The quadratic covariation terms are:

$$(\mathrm{d}r_D)^2 = \sigma_D^2 r_D^2\,\mathrm{d}t, \quad \mathrm{d}r_D\,\mathrm{d}w = \rho\,\sigma_D\sigma_w r_D\,w\,\mathrm{d}t, \quad (\mathrm{d}w)^2 = \sigma_w^2 w^2\,\mathrm{d}t.$$

Substituting into (10) and collecting drift and diffusion components yields (9) after replacing f by r_I via (8). □

Corollary 3.3 (Stochastic Weight as a Function of Time)**.** *Under Definition 3.1, $w(t)$ is itself a stochastic function of time given explicitly by (7), and all three quantities $r_D(t)$, $r_I(t)$, $w(t)$ are consistent with the blending equation (1) for all $t \geq 0$ almost surely.*

4 Analytical Properties of $w(t)$

4.1 Moments and Log-Normality

Since $w(t)$ is a GBM, it is log-normally distributed for every $t > 0$.

Proposition 4.1 (Moments of $w(t)$)**.** *For $n \in \mathbb{N}$:*

$$\mathbb{E}[w(t)^n] = w_0^n \exp\left[n\mu_w t + \tfrac{1}{2}n^2\sigma_w^2 t\right]. \tag{11}$$

In particular,

$$\mathbb{E}[w(t)] = w_0 e^{\mu_w t}, \qquad \mathrm{Var}(w(t)) = w_0^2 e^{2\mu_w t}\left(e^{\sigma_w^2 t} - 1\right).$$

4.2 Singularity Set

Define the *blow-up set*

$$\mathcal{S} := \{t \geq 0 : 2r_D(t) = r_I(t)\}.$$

Since $r_D(t)$ follows GBM and $w(t)$ is a separate GBM, the condition $2r_D(t) = r_I(t)$ translates, via (8), to $w(t) = \infty$, which is a null event under $\mathbb{P}$ for any finite t. Hence $\mathcal{S}$ is a $\mathbb{P}$-null set and $w(t)$ is finite almost surely for all $t \geq 0$.

4.3 Correlation Structure

Proposition 4.2 (Instantaneous Correlation)**.** *The instantaneous correlation between $\mathrm{d}r_D$ and $\mathrm{d}r_I$ is*

$$\mathrm{corr}(\mathrm{d}r_D, \mathrm{d}r_I) = \frac{\sigma_D(2 - w^{-1})^2 + \rho\,\sigma_w w^{-1}(2 - w^{-1})}{\sqrt{[\sigma_D^2(2 - w^{-1})^2 + \sigma_w^2 w^{-2} + 2\rho\,\sigma_D\sigma_w(2 - w^{-1})w^{-1}]\,\sigma_D^2}}. \tag{12}$$

5 Economic and Financial Interpretations

5.1 Connection to Uncovered Interest Parity

Under Uncovered Interest Parity (UIP), the expected depreciation of the domestic currency equals the interest rate differential [3, 4]:

$$\mathbb{E}_t\left[\frac{S(t+1)}{S(t)}\right] - 1 = r_I(t) - r_D(t),$$

where $S(t)$ is the spot exchange rate (domestic per foreign unit). The blending equation (1) can be rewritten as

$$r_I = r_D + (1-2w)(r_I - r_D),$$

so $w = 1/2$ corresponds to the case in which the spread receives zero net loading—i.e. the international rate equals the domestic rate, which is the UIP boundary. Deviations of $w(t)$ from $1/2$ therefore measure the *excess weight* placed on the domestic rate relative to UIP.

5.2 No-Arbitrage Constraint

In an arbitrage-free market, the pricing kernel $M(t)$ satisfies

$$\mathbb{E}_t[M(t+\mathrm{d}t)\, r_I(t)] = r_D(t)\,\mathrm{d}t.$$

The stochastic weight $w(t)$ can be interpreted as the *Radon–Nikodým derivative* factor adjusting the risk-neutral measure for the spread term, consistent with the change-of-numeraire technique in interest-rate modelling [5].

5.3 Central Bank Policy Interpretation

Table 2 summarises the policy interpretation of different ranges of $w(t)$.

Table 2: Policy interpretation of the blending weight $w(t)$.

Range of $w(t)$	Interpretation	Policy implication
$w < 0$	Negative loading on r_D	Inverse domestic-rate exposure
$w \in (0, \frac{1}{2})$	Below UIP boundary	Spread-dominated regime
$w = \frac{1}{2}$	UIP boundary	Pure spread parity
$w \in (\frac{1}{2}, 1)$	Above UIP boundary	Domestic-rate-dominated
$w = 1$	Full domestic loading	$r_I = r_D$ (closed economy)
$w > 1$	Leveraged domestic exposure	Amplified rate pass-through

6 Machine Learning and Data Science Connections

6.1 Latent Factor Representation

The blending equation (1) can be cast as a *single hidden layer neural network* with activation ϕ:

$$r_I(t) = w(t)\,\phi_1(r_D(t)) + \big(1 - w(t)\big)\,\phi_2(r_D(t), r_I(t))\,,$$

where $\phi_1(x) = x$ and $\phi_2(x, y) = y - x$ are the two "neurons." The weight $w(t)$ then plays the role of a *gating function*, analogous to the gate mechanism in an LSTM or GRU cell [7].

6.2 Gaussian Process Prior on $w(t)$

Instead of specifying a parametric GBM for $w(t)$, one may place a Gaussian Process (GP) prior:

$$w(\cdot) \sim \mathcal{GP}(m(\cdot),\, k(\cdot,\cdot))\,,$$

where a stationary Matérn-$\frac{5}{2}$ kernel is a natural choice [8]:

$$k(t,t') = \left(1 + \frac{\sqrt{5}\,|t-t'|}{\ell} + \frac{5(t-t')^2}{3\ell^2}\right)\exp\left(-\frac{\sqrt{5}\,|t-t'|}{\ell}\right).$$

Posterior inference over $w(t)$ given observations of r_D and r_I yields a tractable Bayesian estimator for the blending weight.

6.3 Neural SDE Calibration

A *Neural SDE* [9] replaces the constant GBM coefficients with neural networks:

$$\mathrm{d}r_D(t) = \mu_\theta(r_D, t)\,\mathrm{d}t + \sigma_\theta(r_D, t)\,\mathrm{d}B_1, \tag{13}$$

$$\mathrm{d}w(t) = \tilde{\mu}_\phi(w, t)\,\mathrm{d}t + \tilde{\sigma}_\phi(w, t)\,\mathrm{d}B_2, \tag{14}$$

where θ, ϕ are learnable parameters optimised by minimising the Wasserstein-2 distance between the model-implied distribution of $r_I(t)$ and the empirical distribution from market data.

Table 3 compares modelling approaches.

Table 3: Comparison of modelling approaches for $w(t)$.

Model	Parameters	Tractability	Flexibility	Interpretable
GBM (this paper)	μ_w, σ_w, ρ	Closed form	Low	High
Ornstein–Uhlenbeck	κ, θ, σ	Closed form	Medium	High
Gaussian Process	ℓ, kernel type	Exact (small N)	High	Medium
Neural SDE	θ, ϕ (millions)	Simulation	Very high	Low

7 Simulation Algorithm

Algorithm 1 describes an Euler–Maruyama Monte Carlo procedure for jointly simulating $r_D(t)$, $w(t)$, and the derived $r_I(t)$.

Algorithm 1 Monte Carlo Simulation of $(r_D(t), w(t), r_I(t))$

Require: Parameters $\mu_D, \sigma_D, \mu_w, \sigma_w, \rho$; initial values r_0, w_0; time horizon T; steps N; paths M.
Ensure: Arrays $\hat{r}_D, \hat{w}, \hat{r}_I \in \mathbb{R}^{M\times(N+1)}$.

```
Δt ← T/N
L ← [σ_D  0; ρ σ_w  σ_w √(1 − ρ²)]          ▷ Cholesky factor of correlation matrix
for m = 1, ..., M do
    r̂_D[m, 0] ← r_0;   ŵ[m, 0] ← w_0
    for n = 0, ..., N − 1 do
        Draw Z ~ N(0, I_2)
        ξ ← L Z √Δt                          ▷ Correlated Brownian increments
        r̂_D[m, n+1] ← r̂_D[m, n] exp[(μ_D − ½ σ_D²) Δt + ξ_1]
        ŵ[m, n+1] ← ŵ[m, n] exp[(μ_w − ½ σ_w²) Δt + ξ_2]
        r̂_I[m, n+1] ← r̂_D[m, n+1] (2 − 1/ŵ[m, n+1])
    end for
end for
return r̂_D, ŵ, r̂_I
```

Remark 7.1. Using the exact GBM solutions (6)–(7) for r_D and w in Algorithm 1 eliminates discretisation error for the primitive processes; r_I is then computed from (8) without any Euler approximation.

8 Numerical Results

Table 4 lists the baseline parameter set used in the illustrative computations.

Table 4: Baseline simulation parameters.

Parameter	Description	Baseline	Unit
r_0	Initial domestic rate	0.05	—
w_0	Initial blending weight	0.70	—
μ_D	Drift of r_D	0.02	yr^{-1}
σ_D	Volatility of r_D	0.10	$\text{yr}^{-1/2}$
μ_w	Drift of w	0.00	yr^{-1}
σ_w	Volatility of w	0.15	$\text{yr}^{-1/2}$
ρ	Brownian correlation	0.30	—
T	Time horizon	5.0	yr
M	Monte Carlo paths	10,000	—

Using Proposition 4.1, the theoretical mean and standard deviation of $w(t)$ at $t = 5$ under the baseline are:

$$\mathbb{E}[w(5)] = 0.70\, e^0 = 0.70,$$
$$\text{SD}(w(5)) = 0.70\sqrt{e^{0.15^2\times 5} - 1} \approx 0.70 \times 0.393 = 0.275.$$

9 Illustrations

9.1 System Diagram

Figure 1 shows the causal graph of the model.

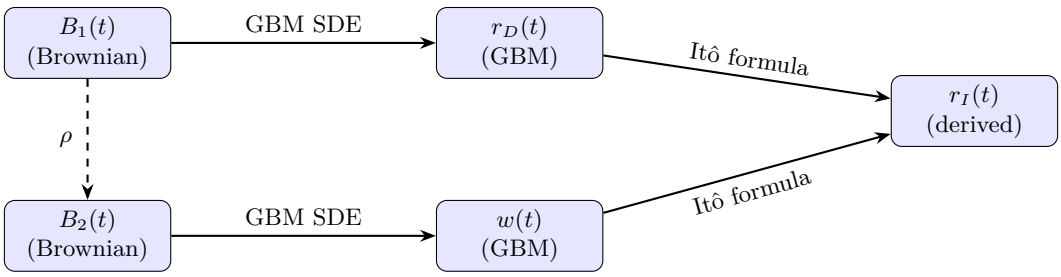

Figure 1: Causal structure: $r_D(t)$ and $w(t)$ are primitive GBM processes driven by correlated Brownian motions; $r_I(t)$ is derived via (8).

9.2 Sample Path Sketch

Figure 2 provides a schematic of typical sample paths.

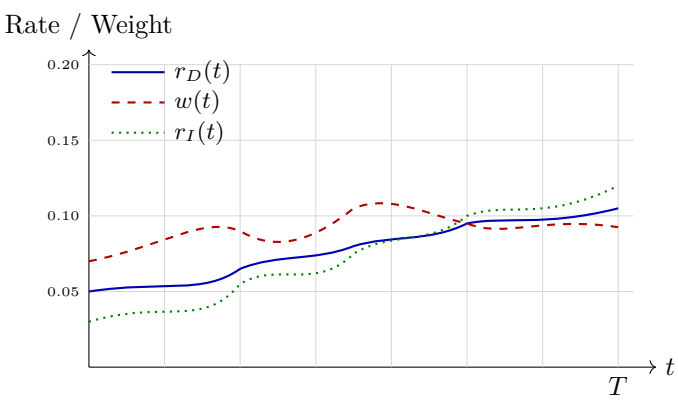

Figure 2: Schematic sample paths of $r_D(t)$ (solid blue), $w(t)$ (dashed red), and the derived $r_I(t)$ (dotted green) over a 5-year horizon.

9.3 Singularity Surface

Figure 3 depicts the region in the (r_D, r_I)-plane where w takes extreme values.

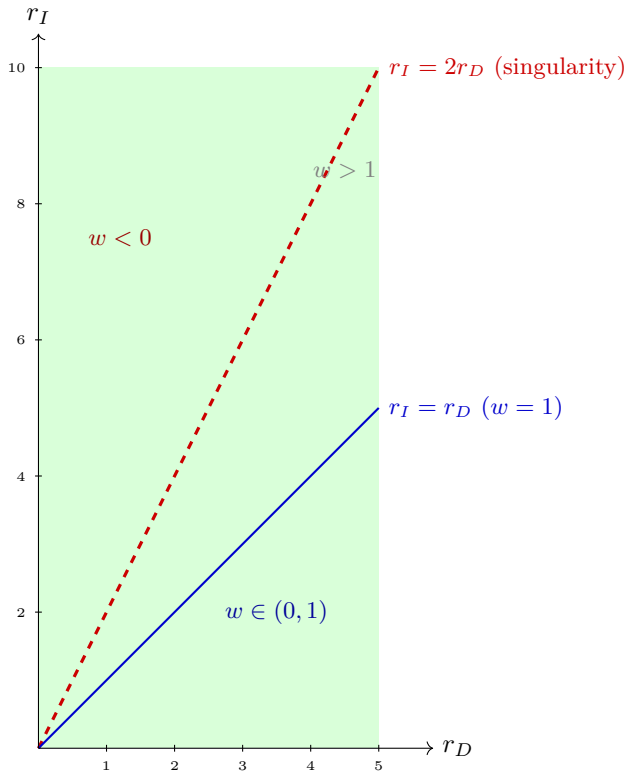

Figure 3: Sign and magnitude regions of $w = r_D/(2r_D - r_I)$ in the (r_D, r_I)-plane. The dashed red line $r_I = 2r_D$ is the singularity; the solid blue line $r_I = r_D$ corresponds to a closed-economy $(w = 1)$.

10 Conclusion

We have shown that the blending weight w implied by the interest rate equation (1) admits the closed-form expression $w = r_D/(2r_D - r_I)$. When r_D and w are modelled as independent (possibly correlated) GBMs, the international rate r_I is a derived process whose SDE follows from Itô's formula and involves both Brownian drivers. The framework accommodates a rich variety of economic regimes (UIP boundary, leveraged pass-through, sign reversal) and connects naturally to modern machine learning calibration methods including Gaussian Processes and Neural SDEs.

Future work should consider: (i) replacing GBM with CIR or Hull–White dynamics to ensure non-negative rates; (ii) empirical calibration using sovereign yield curve data; and (iii) deep hedging strategies exploiting the derived dynamics of $r_I(t)$.

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Glossary

$r_D(t)$ Domestic nominal interest rate at time t; modelled as a Geometric Brownian Motion with drift μ_D and volatility σ_D.

$r_I(t)$ International (foreign) interest rate at time t; derived from $r_D(t)$ and $w(t)$ via the blending equation (8).

$w(t)$ Blending weight; the scalar satisfying the mixture relation (1), equal to $r_D/(2r_D - r_I)$ when rates are deterministic.

GBM Geometric Brownian Motion; a stochastic process $X(t) = X(0)\exp[(\mu - \frac{1}{2}\sigma^2)t + \sigma B(t)]$.

Itô's Formula The stochastic-calculus chain rule for twice continuously differentiable functions of Itô processes, including a correction term proportional to the quadratic covariation.

UIP Uncovered Interest Parity; the condition that expected exchange rate depreciation equals the interest rate differential.

GBM singularity The event $2r_D(t) = r_I(t)$, at which $w(t)$ diverges; this occurs with probability zero for any fixed t.

Euler–Maruyama A first-order time-stepping method for SDEs, analogous to Euler's method for ODEs, in which Brownian increments replace deterministic step multiplications.

Neural SDE A stochastic differential equation whose drift and diffusion coefficients are parameterised by neural networks and calibrated to empirical data.

Gaussian Process (GP) A probability distribution over functions, fully specified by a mean function and a covariance (kernel) function; used here as a non-parametric prior for $w(\cdot)$.

Matérn kernel A family of stationary covariance kernels parameterised by a smoothness index ν and length-scale ℓ; the Matérn-$\frac{5}{2}$ variant yields twice-differentiable sample paths.

Cholesky decomposition Factorisation $\Sigma = LL^\top$ of a positive-definite matrix; used in Algorithm 1 to generate correlated Brownian increments.

Wasserstein-2 distance A metric on probability distributions measuring the minimum transport cost under squared Euclidean ground cost; used as a calibration loss for Neural SDEs.

Radon–Nikodým derivative The density $\mathrm{d}\mathbb{Q}/\mathrm{d}\mathbb{P}$ relating two equivalent probability measures; arises in change-of-numeraire arguments in interest-rate modelling.

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